Work along a displacement

antocciphysics

UNIT 4 / DAY 2 / CPA + SEI

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate work and identify when a force does no work.

SUCCESS CRITERIA - I can...

1. Identify the force component along displacement.
2. Use meters and newtons to obtain joules.
3. Distinguish effort from mechanical work on an object.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “work” and “displacement” to explain your answer. Sentence starters are optional.

work: energy transferred by a force along a displacement

displacement: change from initial position to final position, with direction

parallel: along the same direction as a line or surface

01 Warm-up: bring back an earlier idea

A simplified deflection adds **0.3 m/s** sideways for a lead time of **50 s**. **Find the predicted shift** and **name something you** are treating as true.

Word help: deflection = a change in the direction or path of motion

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: ① ② ③

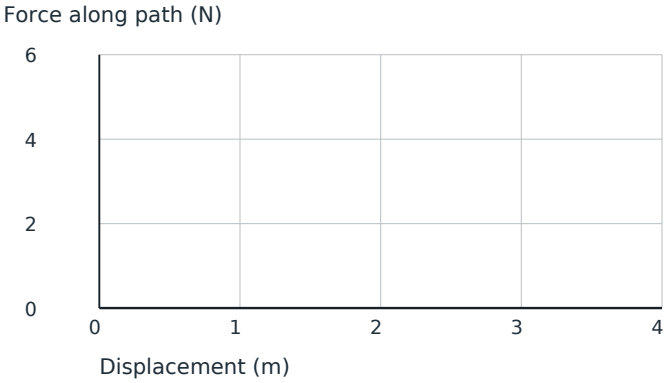
02 Worked example: explain the prediction

Reference: W = Fd.
Constant force parallel to displacement; W is work. No displacement means no work on the object by that force.

Worked reasoning: A constant **5 N** pull acts along a **3 m** displacement. Work is Fd=**15 J**. The object must move for this force to do mechanical work.

Your turn: 1. **Draw the force** and displacement arrows. 2. **Show the 15 J** as area on a force-versus-position graph.

Sentence starter: In my work, I used … to show … . I labeled it … .

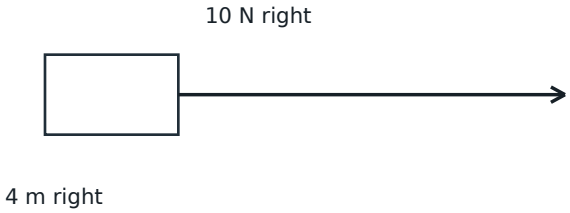


Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A box moves **4 m** right while a **10 N** force acts right. **Draw force** and displacement arrows; **predict the sign of work**.

Sentence starter: I predict … . I used … to make this prediction because … .



The rectangle is the box. Add the requested force and displacement labels.

Marzano check - circle: ① ② ③

Work along a displacement

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

You pull a box with the **10 N** horizontal force shown in question **3**.
1. **Find the work over 4 m**, then over **2 m**.
2. **Draw each force** and displacement; **explain why the shorter trip** changes the work.

Sentence starter: I think … . I can point to …, which shows …, so … .

Add force and displacement arrows; label the angle.

Force: ____ N Displacement: ____ m

Marzano check - circle: 1 2 3

05 Explain your reasoning

A student holds a heavy bag still. Is the upward support force doing mechanical work on the bag during the hold? Explain without denying the student feels tired.

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: 1 2 3

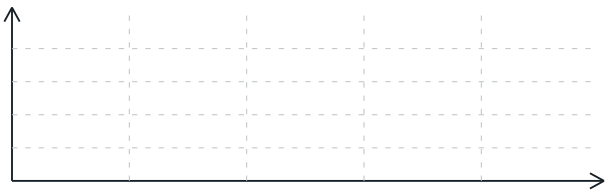
06 Review: connect an earlier idea

The same object stops from the same speed, but the stopping time is **4** times the original. **Compare average stopping-force magnitudes**.
Show the signed impulse on the force–time graph.

Word help: impulse = net force acting over time; change in momentum

Sentence starter: Both … . They differ in … because … .

Force (N)



Time (s)

Marzano check - circle: 1 2 3

07 Exit check

A **3 N** force acts forward through **5 m**. **Find its work**.

Sentence starter: I found … [unit]. This value means … in this situation.

Add force and displacement arrows; label the angle.

Force: ____ N Displacement: ____ m

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

1. Design two force-and-distance pairs doing **20 J** of positive work.
2. **Sketch both**.

Sentence starter: My design includes … . It does what is needed by … .

Add force and displacement arrows; label the angle.

Force: ____ N Displacement: ____ m

Marzano check - circle: 1 2 3